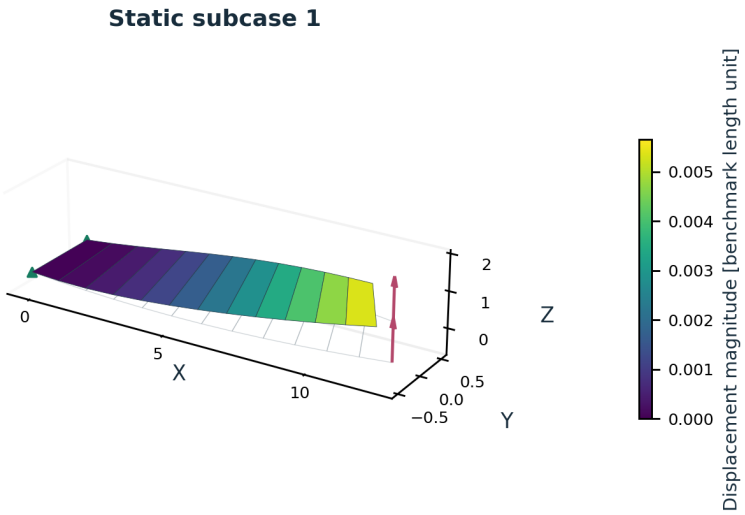
# Twisted cantilever: in-plane tip load

**MH_twisted_beam_in_plane | SOL 101 | Published shell benchmark**

90-degree twisted rectangular cantilever; length 12, width 1.1, thickness 0.32; E=29e6, nu=0.22.

| | |
|---|---|
| Geometry / mesh | 26 nodes; 12 elements. 90-degree twisted rectangular cantilever; length 12, width 1.1, thickness 0.32; E=29e6, nu=0.22. |
| Material / section | E=2.9e+07, nu=0.22; shell thickness=0.32. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2 |
| Boundary conditions | Root clamped; unit total in-plane tip force along global Z. |
| Applied load | Total Fz=1 at the twisted free section (local in-plane bending). |



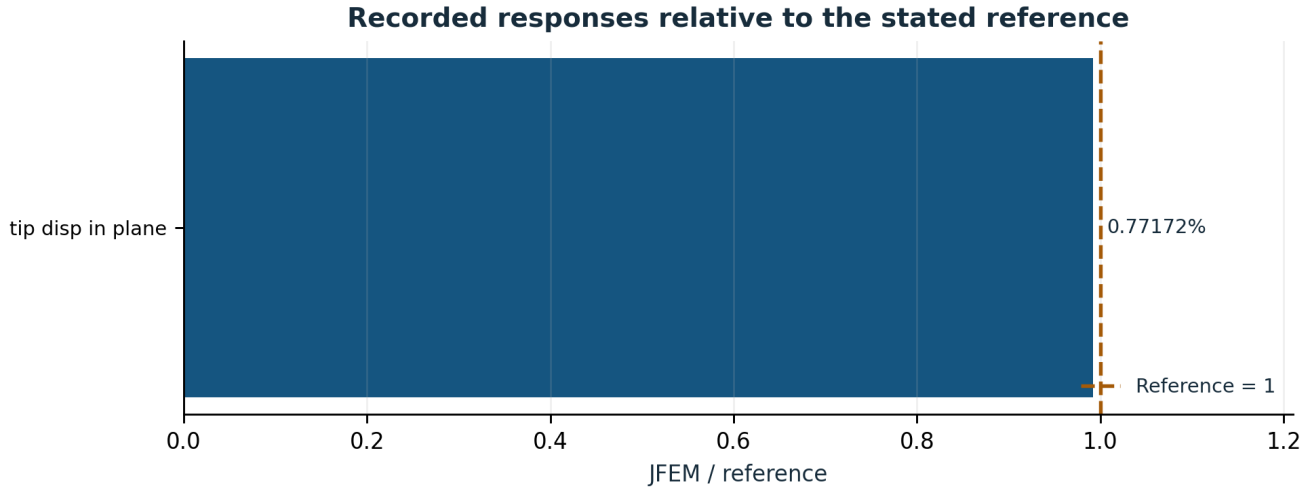

Shape amplified x255; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

## Recorded numerical checks

| Quantity [unit] | JFEM | Reference | Error | Assessment |
|---|---|---|---|---|
| tip disp in plane [benchmark length unit] | 0.0053821 | 0.005424 | 0.77172% | PASS |

# Evidence, convergence and interpretation




Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

## Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv